

LECTURE 1

INTRODUCTION

CEN 348- Artificial Intelligence Systems
Department of Computer Engineering, Cukurova
University

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About the course

- Success grade= 40% of midterm exam grade + 60% of final exam grade.
- Homeworks can be asked for some topics and the homeworks can be added to the midterm/final exam grade.
- Reference book:
 - Artificial Intelligence: A Modern Approach by Stuart J. Russell and Peter Norvig
 - <https://cs.calvin.edu/courses/cs/344/kvlinden/resources/AIMA-3rd-edition.pdf>

Outline of Lecture 1

- A brief introduction to the concept of Artificial Intelligence (AI)
 - What is AI?
 - Basics of AI
 - The gestation of AI
 - Some implementation of AI in the real-world

Artificial Intelligence (AI)

- AI is one of the newest fields in science and engineering.
- Work started in earnest soon after World War II, and the name itself was coined in 1956.
- AI is relevant to any intellectual task; it is truly a universal field.
- AI currently encompasses a huge variety of subfields, ranging from the general (learning and perception) to the specific, such as playing chess, proving mathematical theorems, writing poetry, driving a car on a crowded street, and diagnosing diseases.

What is AI?

- Computer science defines AI research as the study of "intelligent agents": any device that perceives its environment and takes actions that maximize its chance of successfully achieving its goals.
- A more elaborate definition characterizes AI as "a system's ability to correctly interpret external data, to learn from such data, and to use those learnings to achieve specific goals and tasks through flexible adaptation."

Basics of AI

- A typical AI analyzes its environment and takes actions that maximize its chance of success.
- An AI's intended utility function (or goal) can be simple ("1 if the AI wins a game of Go, 0 otherwise") or complex ("Perform actions mathematically similar to ones that succeeded in the past").
- Goals can be explicitly defined or induced.

Basics of AI

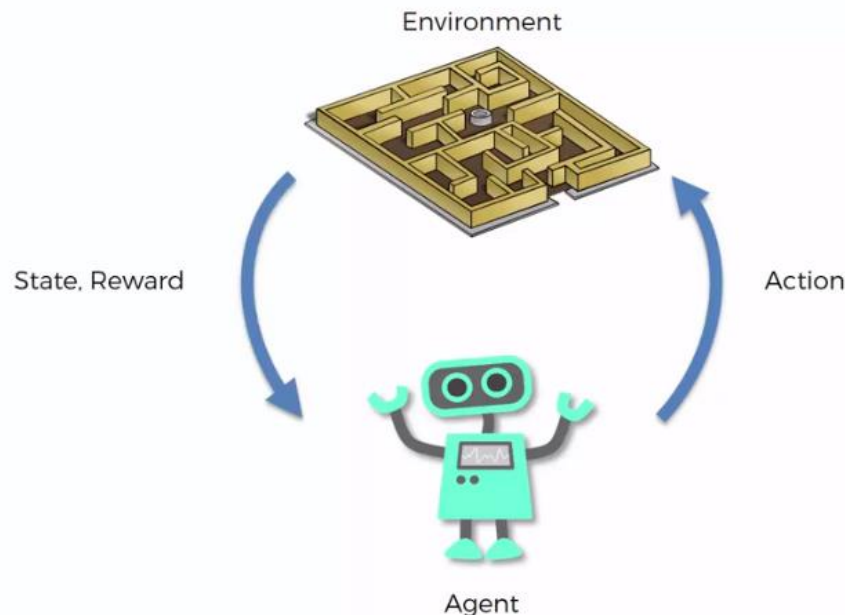
- If the AI is programmed for "**reinforcement learning**", goals can be implicitly induced by rewarding some types of behavior or punishing others.

Examples of Reinforcement Learning:

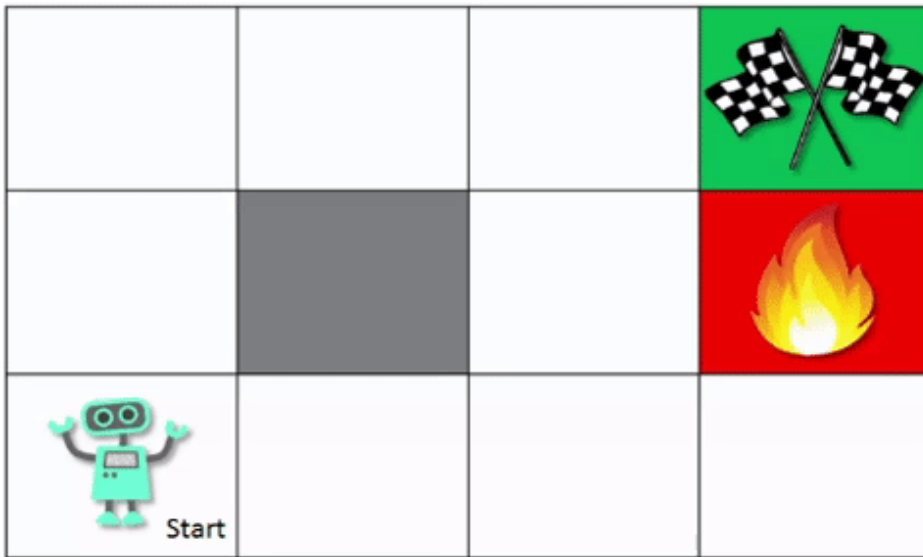
- Learn to play chess
- Making a robot walk
- Fly stunt manoeuvres in helicopter

An example of Reinforcement Learning

- Reinforcement learning has a 'Environment' which is the problem set we are trying to solve and an 'Agent'.
- The relation between Environment and Agent is really simply.
- The Agent will perform certain action and as a result of this action his state will change which will lead to Agent getting a reward.
- By iterating through action and reward process, Agent learns the Environment.
- It comes to understand in which state, which action gives him maximum reward and which give him negative reward.



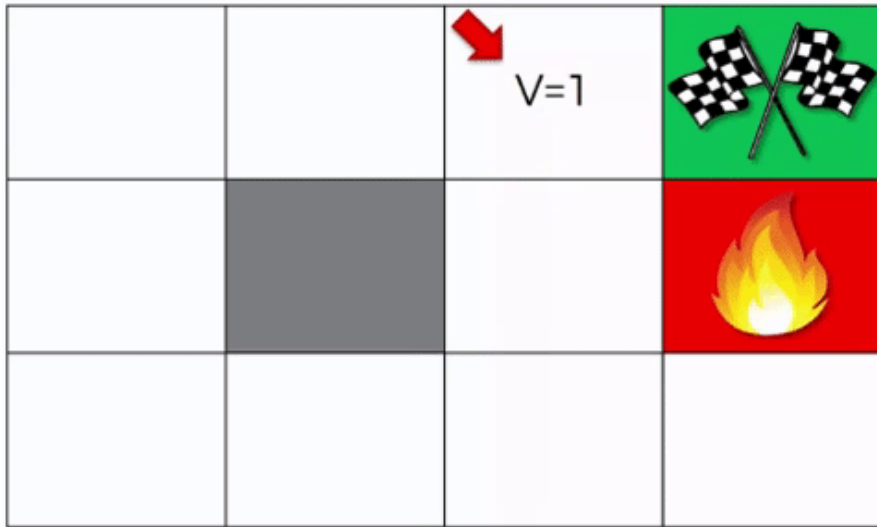
An example of Reinforcement Learning



- The Agent then starts to ask question, how did I get to this square, what was the preceding state I was in and what action I took that got me the reward. It looks back & backtracks its preceding states.

- Let's assume our Agent starts in an environment as shown in 'start' image of adjacent gif.
- Initially it would simply make random movements, like going to the right, left, up, down etc.
- Eventually at some point it would reach its destination.
- As soon as it reaches its destination, it gets a reward, let's say of +1 point.
- This triggers the algorithm to realize that this place (the green square) is where it needs to be.

An example of Reinforcement Learning



- It first marks the square to the left of green square with a virtual reward of '1', telling itself that he just needs to reach that preceding square, and then all it needs to do is take an action of going right and he will reach his destination.

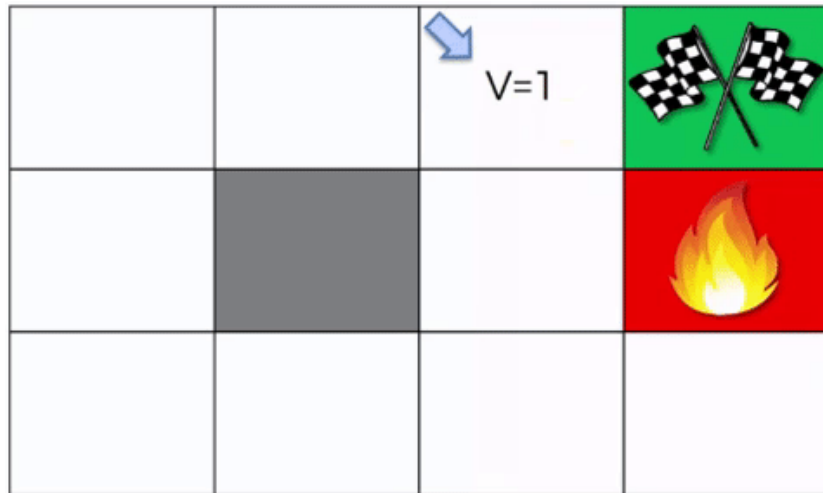
This process continues, till he backtracks his complete path. This backtracking of route, agent performs over multiple iterations.

Now everything would have worked if we would never reset the agent and it will always stay in ON state with the map he just carved out above memorized in its memory.

An example of Reinforcement Learning

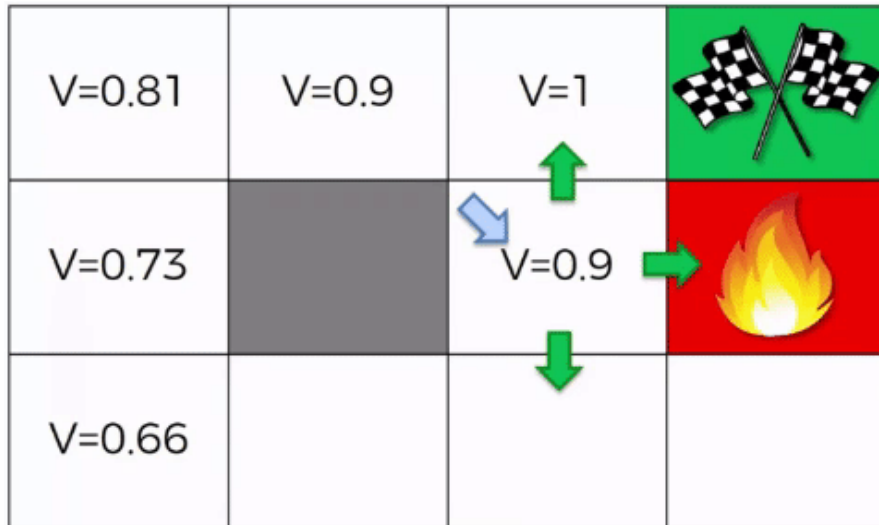
- This is where Bellman equation comes into play.
- Bellman Equation $\rightarrow V(s)$ where $V(s)$ is a value in any given state
- $V(s) = \max(R(s, a) + \gamma V(s'))$
- s — State,
- a — Action,
- R — Reward,
- γ — Discount,
- s' — next state after taking action 'a'

An example of Reinforcement Learning



- The value of the square left to destination Green square is 1, as we can get a reward of 1 in that state if we move to right.
- Backtracking, now instead of assigning the next left square also as 1, as in previous case, we use Bellman equation with a gamma value of 0.9 and assign the next square a value of 0.9 and so and so forth for rest of the route.

An example of Reinforcement Learning



- Link for Bellman equation:
- <https://www.youtube.com/watch?v=14BfO5IMiuk>

- Similarly we can backtrack other route and fill up our map.
- Now that we have created our map, it becomes pretty obvious for our Agent which way it should go no matter which position it starts from.

Basics of AI

- Alternatively, an evolutionary system can induce goals by using a "fitness function" to mutate and preferentially replicate high-scoring AI systems, similar to how animals evolved to innately desire certain goals such as finding food.
- Some AI systems such as nearest-neighbor, these systems are not generally given goals, except to the degree that goals are implicit in their training data.

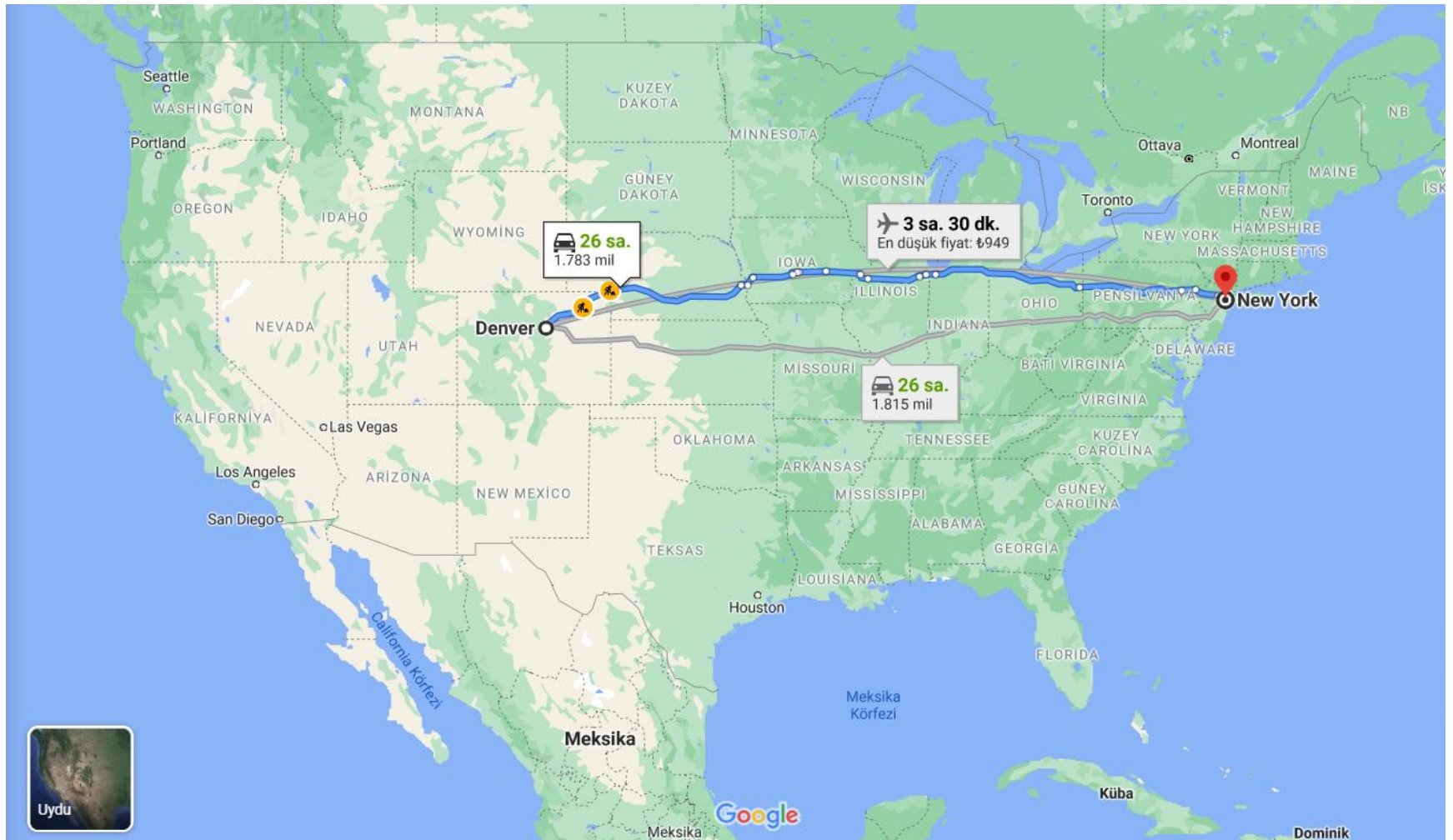
Basics of AI

- Many AI algorithms are capable of learning from data.
- Some of the "learners", including Bayesian networks, decision trees, and nearest-neighbor, could theoretically, (given infinite data, time, and memory) learn to approximate any function, including which combination of mathematical functions would best describe the world.
- These learners could therefore derive all possible knowledge, by considering every possible hypothesis and matching them against the data.
- In practice, it is seldom possible to consider every possibility, because of the phenomenon of "**combinatorial explosion**", where the time needed to solve a problem grows exponentially.

Basics of AI

- Much of AI research involves figuring out how to identify and avoid considering a broad range of possibilities unlikely to be beneficial.
- For example, when viewing a map and looking for the shortest driving route from Denver to New York in the East, one can in most cases skip looking at any path through San Francisco or other areas far to the West.
- Thus, an AI wielding a pathfinding algorithm like **A*** can avoid the combinatorial explosion that would ensue if every possible route had to be ponderously considered.

Basics of AI

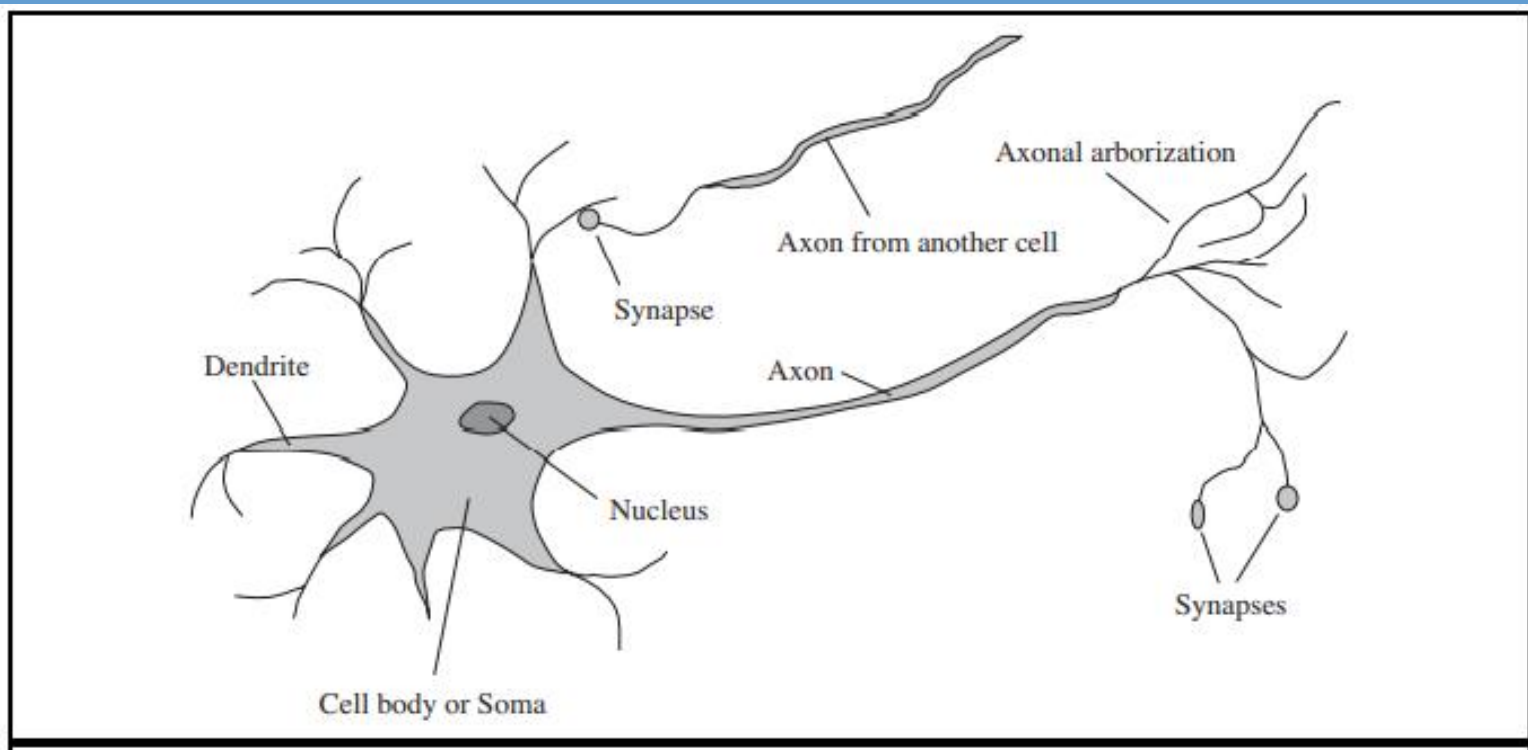


The gestation of AI

- Paul Broca's (1824–1880) study of aphasia (speech deficit) in brain-damaged patients in 1861 demonstrated the existence of localized areas of the brain responsible for specific cognitive functions.
- In particular, he showed that speech production was localized to the portion of the left hemisphere now called Broca's area.
- By that time, it was known that NEURON the brain consisted of nerve cells, or neurons, but it was not until 1873 that Camillo Golgi
- (1843–1926) developed a staining technique allowing the observation of individual neurons in the brain.
- This technique was used by Santiago Ramony Cajal (1852–
- 1934) in his pioneering studies of the brain's neuronal structures.
- **Nicolas Rashevsky (1936,1938) was the first to apply mathematical models to the study of the nervous system.**

The gestation of AI

- The first work that is now generally recognized as AI was done by Warren McCulloch and Walter Pitts (1943).
- They showed, for example, that any computable function could be computed by some network of connected neurons, and that all the logical connectives (and, or, not, etc.) could be implemented by simple net structures.
- McCulloch and Pitts also suggested that suitably defined networks could learn.
- Donald Hebb (1949) demonstrated a simple updating rule for modifying the connection strengths between neurons.
- His rule, now called Hebbian learning, remains an influential model to this day



- The parts of a nerve cell or neuron. Each neuron consists of a cell body, or soma, that contains a cell nucleus.
- Branching out from the cell body are a number of fibers called dendrites and a single long fiber called the axon.
- A neuron makes connections with 10 to 100,000 other neurons at junctions called synapses.
- Signals are propagated from neuron to neuron by a complicated electrochemical reaction.
- The signals control brain activity in the short term and also enable long-term changes in the connectivity of neurons.
- These mechanisms are thought to form the basis for learning in the brain.